BUILD WITH AI · CODE FOR COMMUNITIES | PHC HEALTH SUPPLY CHAIN CHALLENGE | TEAM IDEAFORGE

SWASTHYAGRID

A federated AI control room that forecasts PHC stock-outs and moves medicine before the shelf empties

Do not build India another national database. Federate the forecast: data stays with each state, models travel, and a district officer approves every move.

Running today on 40 simulated facilities in 5 Rajasthan districts. At national scale, base case: 1.36 crore stock-out days a year avoided on 7 emergency medicines.

TEAM IDEAFORGE

Team Members	Institution	Role
Harsh Kawatra	Delhi Technological University	Team Leader
Gursimran Kaur	Guru Gobind Singh Indraprastha University	Member

THE CASE IN FIVE NUMBERS

38,241
PHCs and CHCs run India's frontline care (MoHFW, March 2023)

66 to 80%
of key essential medicines found in stock at surveyed public facilities

58,000
snakebite deaths a year; antivenom only works if the PHC holds it

80%
fewer vaccine stock-outs once eVIN made stock visible (UNDP)

1.2 → 21.0 days
ARV cover at PHC Kota-4 after one approved SwasthyaGrid shipment

Build with AI logo

Team IdeaForge logo

Source: MoHFW Health Dynamics of India 2022-23 | PMC8654139 (Puducherry) | Suraweera et al., eLife 2020 | UNDP eVIN | SwasthyaGrid prototype run log (docs/v2-reports/final-audit.md) | Team IdeaForge impact model

India records PHC stock but nobody forecasts or moves it: SwasthyaGrid federates the forecast across states and removes an estimated 1.36 crore stock-out days a year

SEEN LATE, MOVED NEVER

66 to 80%
of key medicines in stock at surveyed PHCs

58,000 snakebite deaths a year

40% health workers absent (2006)

79.9% CHC specialist posts vacant

WHY THE OBVIOUS ANSWER STALLS

Obvious fix	Why it stalls
One national database	Health is a State subject; 36 States and UTs own the data
Replace e-Aushadhi	States run it today; a rip-out takes years
A dashboard per PHC	Pharmacists work by phone, in Hindi, between patients
AI that auto-orders	No officer signs off; one bad order ends trust

OUR ANSWER: FEDERATE THE FORECAST

1. Decision layer on top of e-Aushadhi, HMIS and eVIN-style feeds, not a new record system
2. Each state's data stays in its own BigQuery; Vertex AI shares model weights, never rows
3. Voice for the PHC, a control room for the district; a named officer approves every move

THE BRIEF, ANSWERED LINE BY LINE

Organiser asks for	SwasthyaGrid today	Status
Live stock, beds, staff	Command Centre: 40 facilities, live stream	LIVE
Demand forecast	Days of cover and 7-day footfall forecast	LIVE
Early stock-out warning	Critical below 3 days, emergency below 5	LIVE
Cross-district moves	Refills live; lateral transfers tested	WIRED
Shared models, all states	Vertex AI federated training per state	PILOT
Voice-first access	Hindi, Hinglish, English; 13 tools	LIVE

WHAT ONE FEDERATED LAYER BUYS

1.36 Cr
stock-out days a year avoided (base)

86.1%
key-medicine availability, from 72.2%

6 weeks
from MoU to a live district

Bear case still avoids 81.5 lakh days a year; bull, 2.17 crore.

Source: MoHFW HDI 2022-23 | PMC8654139 | Chaudhury et al., JEP 2006 | eLife 2020 | Constitution, Seventh Schedule | SwasthyaGrid repository audit (Sep 2026) | Team IdeaForge impact model

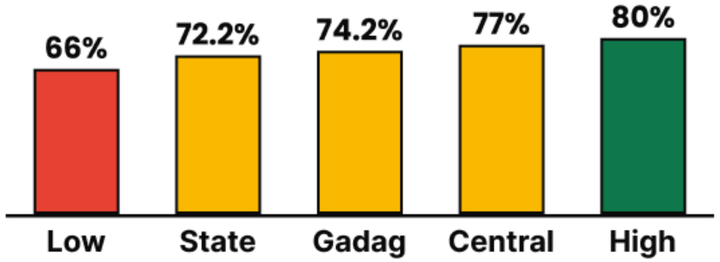
India's 38,241 PHCs and CHCs miss up to 1 in 3 key medicines on a given day, and the emergency drugs that go missing decide whether a patient survives

38,241 PHCS AND CHCS HOLD THE STOCK

Sub-centres	1,69,615
PHCs	31,882
CHCs	6,359
Sub-div. hosp.	1,340
District hosp.	714

Green: the 38,241 facilities we target first

KEY-MEDICINE AVAILABILITY TOPS OUT AT 80%



Key-medicine availability, by study

THE MISSING DRUG DECIDES

Antivenom (ASV)
58,000 snakebite deaths a year

Anti-rabies vaccine
18,000 to 20,000 rabies deaths

Oxytocin
MMR 93; haemorrhage leads

ORS, paracetamol
Dengue signal: 4 to 6 weeks ahead

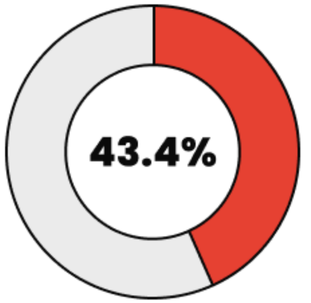
WHY THE SHELF EMPTIES: FIVE WHYS, ONE ROOT

ROOT CAUSE
Stock is recorded but never turned into a decision

- Seen late: HMIS reports monthly, after the shelf empties
- No forecast: nobody computes burn rates or surges
- Not moved: a nearby PHC holds surplus nobody sees
- No hands: 40% of staff absent; registers on paper
- Wrong tool: HQ dashboards, no Hindi voice line

Haryana: 60% of essential medicines were out of stock for 3 to 6 months at a time

HOUSEHOLDS PAY 43.4% OF THE BILL



Out-of-pocket share of total health spending, 2022-23

Medicines are 30.84% of current health spending (2021-22)

An empty PHC shelf sends the family to a private chemist

Source: MoHFW HDI 2022-23 | PMC8654139 (Puducherry) | PMC10292169 (Gadag) | PMC11174260 (Haryana) | eLife 2020 | WHO rabies | SRS 2019-21 | IDSP dengue analysis, PMC13479492 | NHA 2021-22 and 2022-23

India already built the ledgers: e-Aushadhi, HMIS and eVIN record stock, but only eVIN made it visible in real time, and that cut vaccine stock-outs by 80%

FIVE SYSTEMS, ONE MISSING CAPABILITY: DECIDE

System	Record	Live	Forecast	Warn	Move	Voice	Learn
HMIS (monthly)	Yes	No	No	No	No	No	No
e-Aushadhi	Yes	Part	No	Part	Part	No	No
eVIN (vaccines)	Yes	Yes	Part	Yes	Part	No	No
IHIP / IDSP	Part	Yes	Part	Yes	No	No	No
ABDM HFR	Part	No	No	No	No	No	No
SwasthyaGrid	Yes	Yes	Yes	Yes	Yes	Yes	Pilot

Team assessment from public programme documents; eVIN covers vaccines only

THE PRECEDENT: EVIN CUT STOCK-OUTS 80%

80% fewer vaccine stock-outs; restock time more than halved

29,000+ cold chain points live in 733 districts

12 → 36 states at the 2015 launch, then all 36 States and UTs

Same pattern, now for medicines, beds and staff

ONE NATIONAL PLATFORM VS A FEDERATED LAYER

Test	One national platform	Federated layer
Data ownership	State data moves to Delhi	Stays in each state's cloud
Time to pilot	Tender and migration: years	Reads current exports: 6 weeks
Blast radius	One outage stops 36 States/UTs	One state node fails alone
Model quality	One model, one national average	State models, shared weights

FEDERATION IS THE LAWFUL DEFAULT

1. Health is a State subject (Seventh Schedule, List II, Entry 6)
2. DPDP Act 2023: stock data carries no patient identity
3. Google Cloud Mumbai and Delhi regions keep each node in India

Source: C-DAC e-Aushadhi / DVDMS | HMIS, data.gov.in | UNDP India eVIN and NHP eVIN page | IHIP / IDSP | ABDM Health Facility Registry | Constitution, Seventh Schedule | DPDP Act 2023 | Team IdeaForge assessment

SwasthyaGrid closes the loop from shelf to truck: it forecasts each stock-out, proposes the move with a reason, and a district officer approves before anything ships

SEVEN STEPS, ONE LOOP, A HUMAN AT STEP FOUR

1 SENSE
PHC updates stock in the CRM or by voice; Firestore holds it

2 FORECAST
Days of cover, 7-day footfall and bed load per facility

3 PROPOSE
Refill or transfer, with quantity, reason and confidence

4 APPROVE
A district officer approves, edits or rejects

5 DISPATCH
Planner picks the vehicle, driver and cold chain

6 CONFIRM
PHC confirms receipt; stock updates once, idempotent

7 LEARN
Risk drops live; outcomes feed the next forecast

Every confirmed delivery closes the loop: risk falls on the dashboard without a reload

FIVE SURFACES, ONE DATA SPINE

Surface	What it does	Main user
Command Centre	KPIs, risk map, league table, AI briefing	District officers
Voice Control	Hindi, Hinglish, English, grounded in data	PHC staff, MOs
Supply Chain	Dispatch map, fleet, load plan, warehouses	Warehouse teams
Intake CRM	Stock updates and proof of delivery	Pharmacists
Citizen chat	Nearest PHC on Maps; 108 first-aid steps	Patients

ONE APPROVAL MOVED PHC KOTA-4 OUT OF CRITICAL

1.2 → 21.0 days

ARV cover at PHC Kota-4 before and after one approved shipment

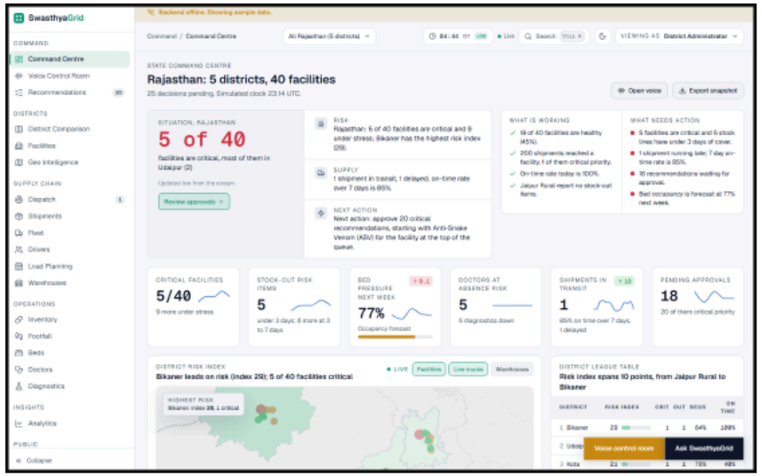
1. 7 vials on the shelf: status critical
2. 119 vials left DDW Kota in a refrigerated van
3. 126 vials confirmed in the CRM: status moves to stress

Simulated operations on real OpenStreetMap road routes

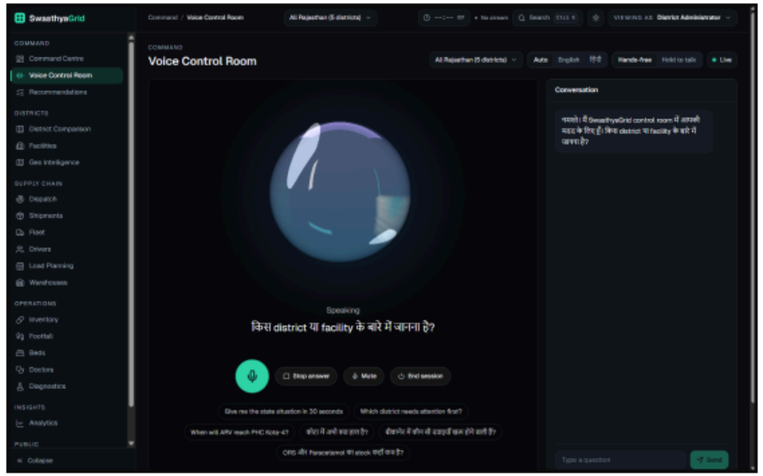
Source: SwasthyaGrid repository (README, docs/v2-architecture.md, backend/app/services) | prototype run log, docs/v2-reports/final-audit.md | OpenStreetMap via OSRM | geoBoundaries

The prototype runs end to end today: 3 live surfaces, 48 API paths with a worst p95 of 65 ms, and 642 automated tests across backend, dashboard and CRM

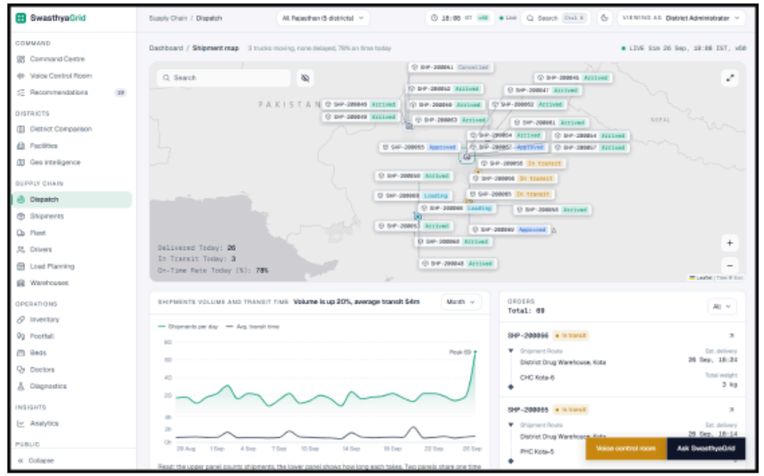
THREE SURFACES, ONE LIVE LOOP



Command Centre: 40 facilities, live risk



Voice Control Room: Hindi and English



Dispatch: trucks on real road routes

642 TESTS AND A 65 MS WORST P95

642	automated tests pass	**37 / 40**	voice eval cases
48	API paths	**836 ms**	median to first audio
65 ms	worst p95, 35 GETs	**365**	real road routes
13	read-only voice tools	**31**	trucks, 2 refrigerated

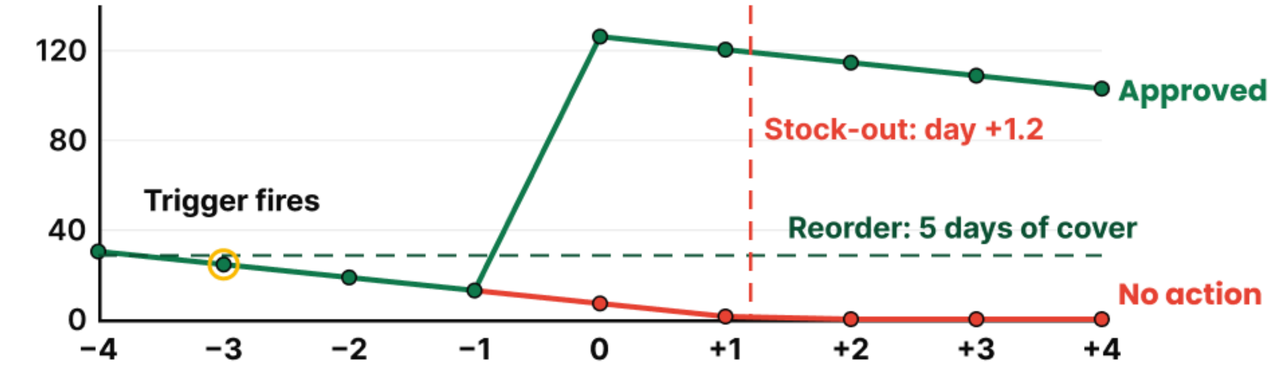
RUNNING ON GOOGLE TODAY

LIVE	Cloud Run in Mumbai (asia-south1): API, live stream, voice socket
LIVE	Gemini API: typed Ask agent and citizen chat on shared tools
LIVE	Firestore and Firebase Admin: CRM intake and proof of delivery
LIVE	Maps Places API: nearest PHC for a citizen

Source: SwasthyaGrid repository, README and docs/v2-reports/final-audit.md (Sep 2026) | swasthyagrid.vercel.app | Cloud Run asia-south1 | github.com/harshkawatra11/SwasthyaGrid-HealthTech

Engines compute every number and Gemini only explains it: each stock-out warning carries days of cover, a stated reason and a confidence score from 40 to 98

REORDER FIRES 4.2 DAYS BEFORE THE SHELF EMPTIES



Burn of 5.8 units a day, from the audited run (7 units = 1.2 days)

ENGINES DECIDE, MODELS PHRASE

Decision	Engine	LLM
Days of cover, risk	Decides	Never
Bed pressure	Decides	Never
Transfer size, priority	Decides	Never
Confidence score	Decides	Never
Approve or reject	A human	Never
Briefings, voice answers	Gives numbers	Phrases them

THE FORMULAS, IN THE OPEN

Days of cover = units ÷ max(daily use, 0.1)

Critical below 3 days; emergency drugs below 5

Confidence = 0.5 × forecast + 25 × (1 − km ÷ 60) + 10 if emergency + 10 if surge, held to 40 to 98

Refill to 21 days; lateral only within 40 km

VERTEX AI TURNS RULES INTO LEARNED FORECASTS

Layer	Today (LIVE)	Pilot on Vertex AI
Demand forecast	Burn-rate rules	AutoML Forecasting per state
Surge signals	Seeded demand factors	IDSP, IMD, Earth Engine floods
Serving	In-process Python	Vertex AI Endpoints, Registry
Cross-state	None	Federated averaging of weights

Source: backend/app/services/forecast_service.py and recommendation_service.py | docs/06-forecasting.md | prototype run log (PHC Kota-4) | Google Cloud Vertex AI documentation | IDSP dengue analysis, PMC13479492

Google AI works in every layer: 8 Google services run in the prototype today, Gemini Live is wired as the voice swap, and 14 more switch on in the pilot

FIVE LAYERS, EVERY SERVICE TAGGED BY STATUS

PEOPLE AND CHANNELS	Sarvam voice: STT, LLM, TTS · Gemini Live API · Dialogflow CX · Cloud Speech-to-Text and TTS · Cloud Translation API
AI AND MODELS	Gemini API · Google AI Studio · Vertex AI: AutoML, training, Endpoints · Gemini multimodal · Vertex AI Vision
DATA AND MAPS	Firebase: Firestore · Firebase Auth · BigQuery · BigQuery Analytics Hub · Maps Places API · Maps Routes API · Earth Engine
RUN AND DEPLOY	Cloud Run (asia-south1) · Secret Manager · Cloud Build · Artifact Registry · Pub/Sub · Cloud Scheduler · Cloud Functions
PUBLIC DATA	data.gov.in HMIS · IDSP / IHIP · IMD weather · ISRO Bhuvan · WHO GHO · OpenStreetMap roads · geoBoundaries

8 LIVE, 1 WIRED, 14 IN THE PILOT

- **LIVE** runs in the prototype today
- **WIRED** code exists, off in the demo
- **PILOT** built in the 6-week pilot

ONE TOOL LAYER, TWO VOICE ENGINES

13 read-only tools in Gemini function schema

Sarvam
LIVE
STT, LLM and TTS

Gemini Live
WIRED
Speech in and out

836 ms to first audio; 37 of 40 eval cases

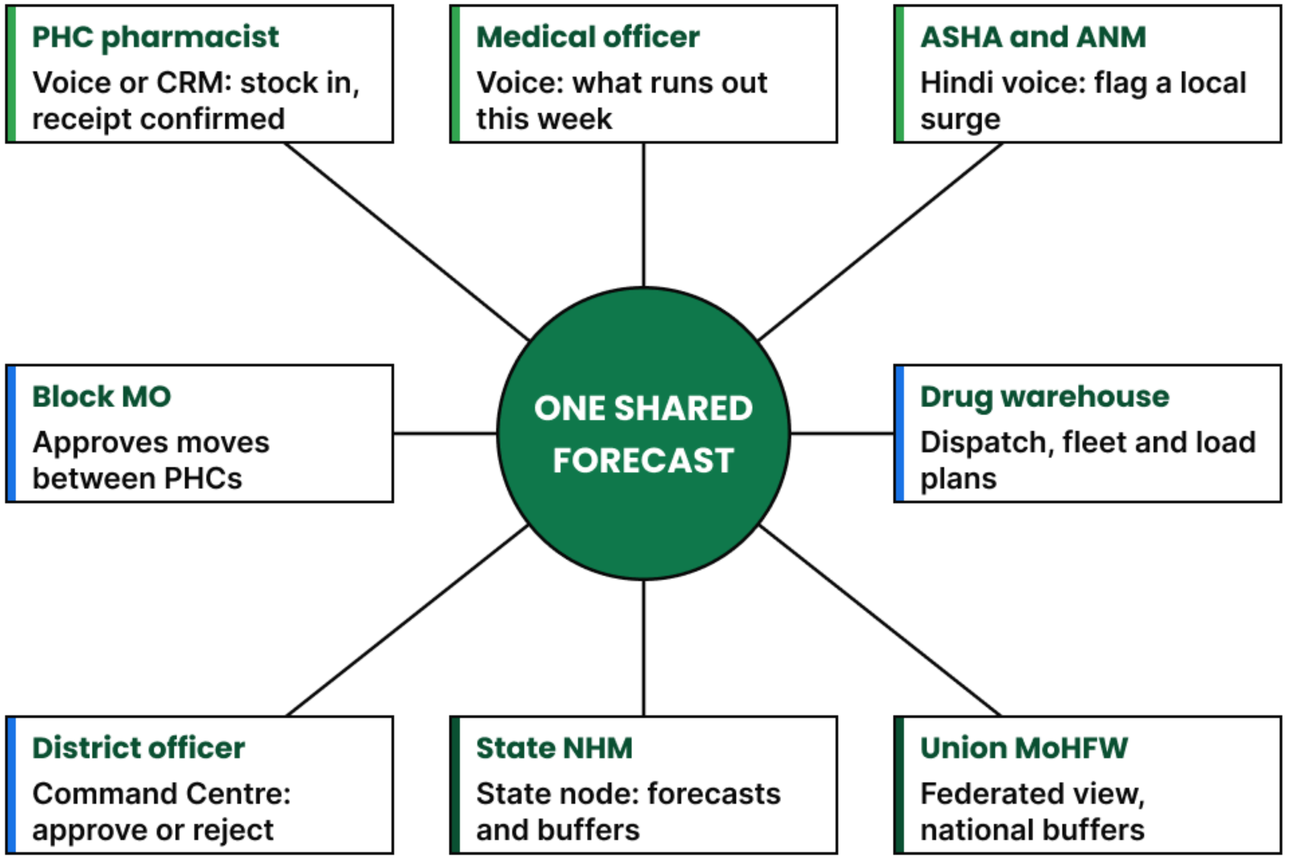
Every number comes from a tool result

Hindi, Hinglish and English today

Source: SwasthyaGrid repository (backend/app, frontend, ai-healthcare-crm, scripts/deploy.sh) | Google Cloud product documentation | data.gov.in | IDSP | IMD | ISRO Bhuvan | WHO GHO | Team IdeaForge build-status audit (Sep 2026)

One forecast serves eight roles from the ASHA worker to the Union ministry, and each gets the interface it can use, in its language, with the decision it owns

EIGHT ROLES AROUND ONE SHARED FORECAST



VOICE FIRST WHERE SCREENS FAIL

Role	Voice	Screen	CRM	Chat
PHC staff	Main	Also	Main	No
ASHA / ANM	Main	No	No	Also
Block, district	Also	Main	No	No
State, Union	No	Main	No	No
Citizens	No	No	No	Main

Live: Hindi, Hinglish, English. Pilot: 22 languages

VALIDATION PLAN: 40 INTERVIEWS, NONE DONE YET

Role	Target	What we test
PHC pharmacists	12	Voice vs CRM for stock
Medical officers	8	Warning lead time needed
District officers	8	Trust in confidence score
Warehouse staff	6	Load plans vs real trucks
State NHM analysts	6	Model governance

Planned before the pilot; no user interviews are claimed

Source: MoHFW Health Dynamics of India 2022-23 (facility tiers and roles) | SwasthyaGrid voice evaluation, docs/v2-reports/final-audit.md | Sarvam AI and Google Cloud Translation documentation | Team IdeaForge validation plan

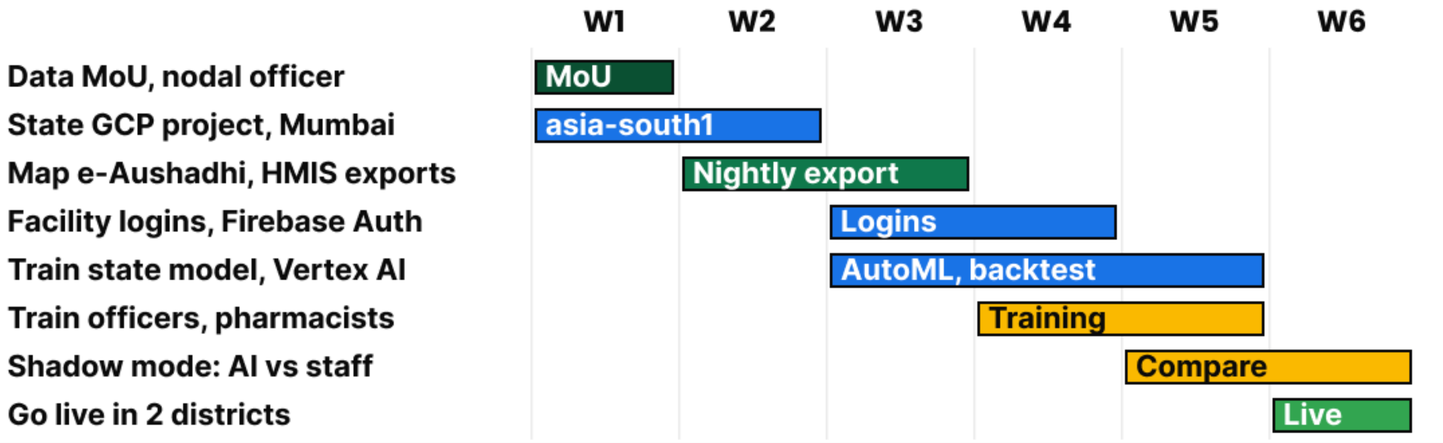
WHY IT'S DEPLOYABLE

DEPLOYABILITY AND SCALABILITY · 20%

9 / 10

A state can go live in six weeks without new hardware: SwasthyaGrid reads the exports states already produce, runs in an Indian Google Cloud region and needs one approver per district

SIX WEEKS FROM MOU TO A LIVE DISTRICT



FITS THE RAILS STATES ALREADY RUN

1. e-Aushadhi / DVDMS: nightly export, no rip-out
2. HMIS on data.gov.in: baseline for backtests
3. ABDM facility registry ID as the facility key
4. Runs on NHM budgets: no new procurement line
5. Officer sign-off mirrors today's indent approval

FIVE RISKS, EACH WITH A FIX

P × I	Low	Med	High
High	R5	R1	
Medium		R3	R2
Low			R4

- R1 Stale data: freshness flag
- R2 Outbreak surge: IDSP signal
- R3 Queue ignored: SLA alerts
- R4 State says no: runs solo
- R5 No signal: offline CRM sync

GUARDRAILS A MINISTRY CAN SIGN

1. AI proposes; a named officer approves
2. Voice agent is read-only by design
3. No patient identity: stock and counts only
4. Every spoken number traced to a tool
5. Data stays in the state's Indian region
6. Runs on the Always Free tier today

Source: SwasthyaGrid deploy script and README (Cloud Run, Secret Manager, Always Free tier) | C-DAC e-Aushadhi / DVDMS | data.gov.in HMIS | ABDM Health Facility Registry | DPDP Act 2023 | Team IdeaForge pilot plan

Rolled out state by state over three years, the federated layer removes an estimated 1.36 crore stock-out days a year and lifts key-medicine availability from 72.2% to 86.1%

1.36 crore
stock-out days avoided a year, base case

72.2% → 86.1%
key-medicine availability

38,241
PHCs and CHCs

36
States and UTs, data kept home

MODELS MOVE, DATA STAYS HOME

Union node: MoHFW
Averages weights; sees national buffers

weights and aggregates only

Rajasthan	Bihar	+ 34 more
BigQuery, Vertex AI	BigQuery, Vertex AI	BigQuery, Vertex AI

Facilities: PHCs, CHCs, warehouses
CRM, voice and photos feed the state node

THREE WAVES TO 36 STATES AND UTS

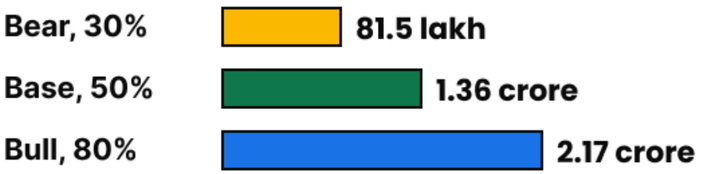
Wave 1 · months 0 to 6
Rajasthan: 2 pilot districts, then statewide

Wave 2 · months 6 to 18
8 states with 70% of snakebite deaths, Bihar to Gujarat

Wave 3 · months 18 to 36
All 36 States and UTs; 38,241 facilities

eVIN scaled from 12 states to all 36

BEAR 81.5 LAKH, BULL 2.17 CRORE



Stock-out days avoided a year, by cut

ALL NINE CASES CLEAR 58 LAKH DAYS

Avail.	30% cut	50% cut	80% cut
66%	99.7	166.1	265.8
72.2%	81.5	135.8	217.3
80%	58.6	97.7	156.3

Source: Team IdeaForge impact model: 38,241 facilities (MoHFW HDI 2022-23) × 7 tracer medicines × 365 days × 27.8% unavailability (PMC8654139); cuts of 30%, 50%, 80% (UNDP eVIN) | eLife 2020

THANK YOU

SwasthyaGrid: federate the forecast, keep each state's data at home, and let a named officer approve every move

swasthyagrid.vercel.app | github.com/harshkawatra11/SwasthyaGrid-HealthTech

TEAM IDEAFORGE

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Gursimran Kaur	Guru Gobind Singh Indraprastha University	Member